

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 268

THE QUINN TEN-RING STAIR

ten graduated 6m rings, 3-second dwell
17.3 RPM at 3m — 19.5 km/h rim



sessional gravity — the fluid column down, the bones loaded

THE ARITHMETIC CLOSED SITTING 121 · PROMOTED SITTING 122

A TREADMILL FOR THE SKELETON

Crew health — v1.0 in force

GP-IM-268

Revision v1.0 — 24 August 2026

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SEATED LAW — EXTERNAL VET PENDING

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 268

PHASE 268: THE QUINN TEN-RING STAIR — STEP UP TO 1G (TEN GRADUATED 6M RINGS, THE 3-SECOND DWELL ON-RAMP; THE VARIABLE-G TREADMILL; THE AUTHOR'S SANS FIX; MEETINGS ON THE WIDE WHEEL) — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING (PROMOTED FROM CAVEAT, SITTING 122; AWAITING BENCH). Sessional, build-now artificial gravity for crew health: the fluid column pulled back down, the eyeballs relieved, the bones loaded — a stair of ten graduated 6 m rings, each 1 m wide, stepped in rate so the crew climbs the g-load like a treadmill. The author's fix for the movie-ring problem, seated in his own words: 'basic movie gravity are city sized rings spinning at 1G with dome covers and 14psi air pressure, logical physics but not b...' — the Ten-Ring Stair answers with no pressure shell, no city, no dome: just rings you stand in. The arithmetic is closed (sitting 121): 1G at 3 m radius is 17.3 RPM, rim speed 5.42 m/s — 19.5 km/h, a brisk bicycle. The caveat stands honest: 17.3 RPM exceeds the ~10 RPM comfort line and the Coriolis at head height runs about 0.37 g — so the doctrine is sessional: minutes on the stair, not hours of habitation; a variable-g treadmill for the skeleton, promoted from caveat to phase in sitting 122, awaiting its bench.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-268, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the stair law as seated (contents line, the stair in crayon, the arithmetic closed, the caveat, the bench items) — §2 the physics, graded — §3 the system on the stair — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 268: **The Ten-Ring Stair — Step Up to 1G (ten graduated 6m rings, the 3-second dwell on-ramp; the variable-g treadmill; the author's SANS fix; meetings on the**

wide wheel).'

THE CORE OBJECTIVE (verbatim): 'Core Objective: sessional, build-now artificial gravity for crew health — the fluid column pulled back down, the eyeballs relieved, the bones loaded — without city-sized movie rings and without the pinball entry that made small rings unusable. Ten rings, ten speeds, one stair.'

Lineage: the author's answer to the roadblock sweep's SANS candidate (sittings 117-120) and the ship's sessional gravity — dictated in sitting 121 (9 August 2026), arithmetic closed by the audit in the same sitting, promotion ordered by the author in sitting 122 ("add that as next phase").

Live Instruction Confirmed: Yes — 9 August 2026 (sittings 121-122).

THE PROBLEM AS GIVEN (VERBATIM): "basic movie gravity are city sized rings spinning at 1G with dome covers and 14psi air pressure, logical physics but not build now and not fit on a ship, a gravity ring even 20 feet high wouldnt work, start up and slow down are momentum points and the pinball effect as is trying to jump onto a ring doing 1g"

THE MACHINE (VERBATIM): "ten rings side by side 6m (2 pod) height each ring spins at different speeds each is a metre wide, the first ring is almost not noticeable, graduations divided evenly by ten . / you step on, zero g you dont fall you are just with the ring wait 3 seconds walk onto the next ring wait 3 seconds repeat until you are on the last ring the variance is so small you barley need to step forward as 1 G is quite slow, once you get on the last ring, you can walk or just stand"

THE ILLUSION FIX AND THE USE DOCTRINE (VERBATIM): "the treadmill thing is an illusion to the brain, you can walk in either direction, to fix that if there was a rotating screen locked to identify tags that keeps level with the walker they would see themselves as running on flat ground or at least just up and down low hills... / in 1G as here, you carry a weighted backpack you feel it the same, so 30 mins minimum per day per person, 10-20 units for our ship, take a chair have a seat read a book, a 3 metered wide last wheel could have long meetings, killing two birds one stone, table legs would be odd but easily done." [AMENDED 12 August 2026 — THE AUTHOR'S RULING: "Amending the staircase crew times, if blood rises, then intestinal contents and subsequent fluid swap from the large bowel will suffer the same issues. amend to 20min twice daily, at least 6 hours apart, pref 8 to 12 hours" — the crew prescription is now 20 MINUTES TWICE DAILY, SESSIONS AT LEAST 6 HOURS APART, PREFERABLY 8 TO 12, superseding the '30 mins minimum per day' seated above (text preserved verbatim). One long pull becomes two spaced pulls: the gut column re-settles twice a day. The audit's grade: direction affirmed (documented crew medicine: fluid shift and slowed gut motility ride together in freefall); the large-bowel fluid-swap extension is the Author's own inference — plausible, consistent, seated as his. Flag for the ledger: per-person demand rises 30 to 40 minutes daily; the 10-20 unit fleet count runs a third tighter — flagged, not fixed.]

THE ARITHMETIC (AUDIT, SITTING 121 — CLOSING IN FULL): 1G at 3-metre radius: $\omega = \sqrt{g/r} = 1.81 \text{ rad/s} = 17.3 \text{ RPM}$; rim speed 5.42 m/s = 19.5 km/h ("1G is quite slow" — verified for the rim). Ten rings graduated evenly in speed: adjacent-step differential a constant 0.54 m/s (2 km/h), held 3 seconds per ring — the pinball entry (a 19.5 km/h instantaneous change) converted into ten 2 km/h steps with vestibular rest between. THE INVENTION IS THE ON-RAMP: the ring was never the problem; the entry

was. G-ladder (linear speed graduation) rises quadratically: 0.01, 0.04, 0.09, 0.16, 0.25, 0.36, 0.49, 0.64, 0.81, 1.00 g — gentlest where a zero-G body is most sensitive. THE VARIABLE-G TREADMILL: walk 1 m/s spinward on the last ring = 1.4 g; 1 m/s anti-spinward = 0.67 g; anti-spinward at rim speed = 0 g under a spinning ring — workout load controlled by the walker's own legs. The tag-locked rotating screen answers the optokinetic conflict (visual-vestibular mismatch = the primary nausea driver in rotating environments). Seated use (reading, the meeting wheel) checks out: semicircular canals signal angular acceleration only — at constant 17 RPM they settle into honest 1G.

THE CAVEAT (SEATED PER STANDING LAW): 17.3 RPM exceeds the ~10 RPM unadapted-comfort line — a 1 m/s head translation on the last ring pulls ~0.37 g Coriolis laterally; first-fortnight users will feel head turns. The design carries its own countermeasure: the graduated 3-second dwells ARE progressive vestibular training (the documented path to double-digit-RPM adaptation), and the 30-minute daily prescription keeps exposure sessional, not continuous.

Why it matters, in plain English: every space agency knows crew need gravity and every ship design punts it to “someday, bigger.” The stair says: don't build bigger — build a ramp. SANS, bone loss, muscle loss and fluid shift are all the same wound — the body parked in freefall — and this is the bench-scale, build-now answer that fits a two-pod ceiling. The meeting wheel means the machine earns its floor space twice.

Status: Promoted by the author 9 August 2026 (sitting 122). Arithmetic closed sitting 121. Awaiting bench: ring-speed synchronization tolerances and the screen-tag loop.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PROBLEM IS QUOTED, NOT STRAWMANNED: the author's own words seat the failure of movie gravity — city-sized rings at 1G with dome covers and 14 psi are logical physics but not buildable by anyone alive. The Ten-Ring Stair is the buildable answer: no shell, no city, no dome — rings you stand in, sessionally.

THE ARITHMETIC IS CLOSED (sitting 121): 1G at 3 m radius is 17.3 RPM, rim speed 5.42 m/s — 19.5 km/h, a brisk bicycle. The forces are human-scale; the machine is shed-scale; the math is on the page.

THE GRADUATION IS THE DESIGN: ten rings at stepped rates — the crew starts low and climbs, the g-load dosed like physio. The 3-second dwell at each on-ramp is the transition law: step on, hold, step up.

THE CAVEAT STANDS HONEST AND SEATED: 17.3 RPM exceeds the ~10 RPM comfort line, and head-height Coriolis runs about 0.37 g — this is not a habitat, it is a treadmill. Sessional minutes for fluid column, eyeballs and bones; the doctrine never overclaims.

SESSIONAL IS THE WHOLE POINT: crew health does not need 1G all day — it needs the skeleton loaded and the fluid column pulled down on a schedule. The stair delivers the dose without building the city.

PROMOTED FROM CAVEAT, SITTING 122: the numbers survived the first pass and the author promoted it to a phase — awaiting bench, honestly stated.

§3 — THE SYSTEM ON THE STAIR

- **The rings:** ten, 6 m diameter, each 1 m wide — the stair in section.
- **The rates:** stepped per ring — the g-load climbs like treadmill incline.
- **The dwell:** 3 seconds at each on-ramp — the transition law.
- **The numbers:** 1G at 3 m — 17.3 RPM, 5.42 m/s rim, 19.5 km/h; closed, sitting 121.
- **The caveat:** above the comfort line, 0.37 g Coriolis at the head — sessional, not habitation.
- **The dose:** fluid column down, eyeballs relieved, bones loaded — minutes, on a schedule.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Sitting 121 closed the arithmetic; sitting 122 promoted it — the audit's numbers, the author's promotion, both seated.
- The centrifuge literacy is family heritage: 129/130's cake-oven g-force and 84's drums taught the file to price rotation honestly — the stair spends that literacy on crew health.
- 246's medical band is the doctrine cousin: the stair is crew medicine administered as architecture — prevention by dose, the same philosophy as the envelope.
- The vet coverage, honest: the phase sits in the 261-277 band the external vet never reached — the caveat and the promotion are the file's own sittings, and §5 says so.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Build ring one: 6 m, 1 m wide, the lowest rate — the stair's first tread.
- 2. Step the rates: the ten-ring graduation table benched — the g-dose per tread.
- 3. Engineer the on-ramps: the 3-second dwell, handholds, the step-up sequence.
- 4. Bench the comfort: session length versus the Coriolis load — the schedule the medics sign.
- 5. Write the dose: minutes per session, sessions per day — crew health as a prescription, not a hope.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phases 129/130 (the centrifuge heritage):** the literacy — rotation priced honestly, one family up.
- **Phase 246 (the medical band):** the cousin — crew medicine by dose, worn there, stood in here.
- **Phase 269 (the 5-1 bank):** the physics cousin — rim speed as stored energy there, rim speed as gravity here; the same $E = 1/2 M v^2$ literacy.

§8 — DO-NOT-BUILD

- Do NOT sell it as habitation — sessional only; the comfort line is real and the caveat is seated.
- Do NOT skip the dwell — 3 seconds at each on-ramp; transitions are where falls happen.
- Do NOT build the city — no shell, no dome, no 14 psi; the stair exists because the movie ring is unbuildable.
- Do NOT overclaim the dose — fluid column, eyeballs, bones; the stair is physio, not paradise.
- Do NOT bench it without medics — the session schedule is a prescription; the medics sign it.

§9 — OWED

OWED: the bench, whole — the graduation table (ten rates), the on-ramp engineering, the session schedule versus measured Coriolis tolerance, drive and brake specs, the fail-safe stop. And the external vet, when it lands.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-268, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING (PROMOTED FROM CAVEAT, SITTING 122; AWAITING BENCH), seated law, master v2.1 (numerical print order). Built 24 August 2026, ninth of the 20-chapter 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will

take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 268